

TASK 4

HAWKEYE LAB ANALYSIS REPORT

1. Introduction

The HawkEye Lab on the CyberDefenders platform is a network forensics challenge in which a .pcap file is provided to investigate a stealer malware infection. The goal is to identify intelligence about IOCs and data exfiltration with Wireshark.

2. Objective

- Reconstruct attack timeline.
- Identify key indicators of compromise (IoCs).
- Analyse attacker behaviour and identify potential security weaknesses exploited during the attack.
- Determine methods used by the malware to extract sensitive information from the victim's machine.

3. Tools Used

- Wireshark – analyses network traffic
- CyberChef – decode encoded Base64 data
- Windows Powershell – generates file hash for malware verification
- Iplocation.net – identifies geolocation of IP addresses.

4. Lab Setup

4.1. Login

Created an account in the CyberDefenders platform (<https://cyberdefenders.org/>) and opened the HawkEye Lab.

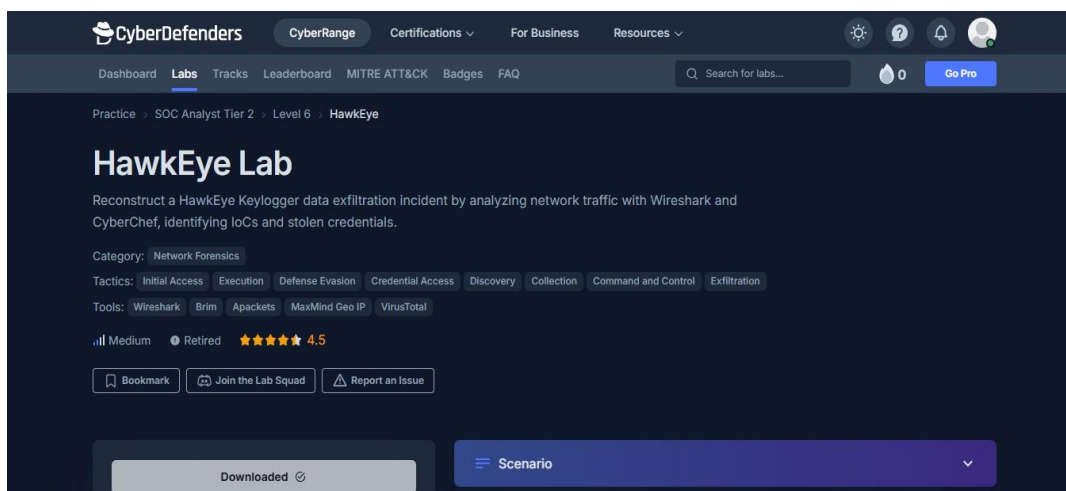


Figure 4.1.

4.2. Dataset Download

The HawkEye Lab files were downloaded from the platform. The dataset had a file named “stealer.pcap”. It was used for analysis to detect suspicious activities.

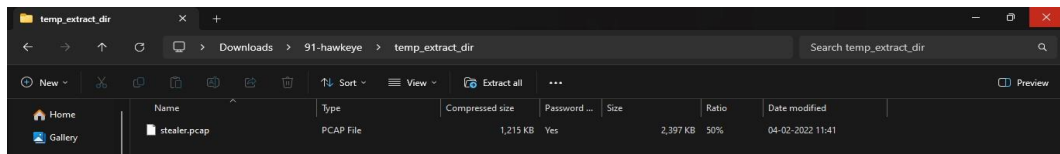


Figure 4.2.

4.3. Tool Installation

Wireshark was installed to analyse network traffic captured in the .pcap file. It helps in identifying suspicious IP addresses, protocols, communication.

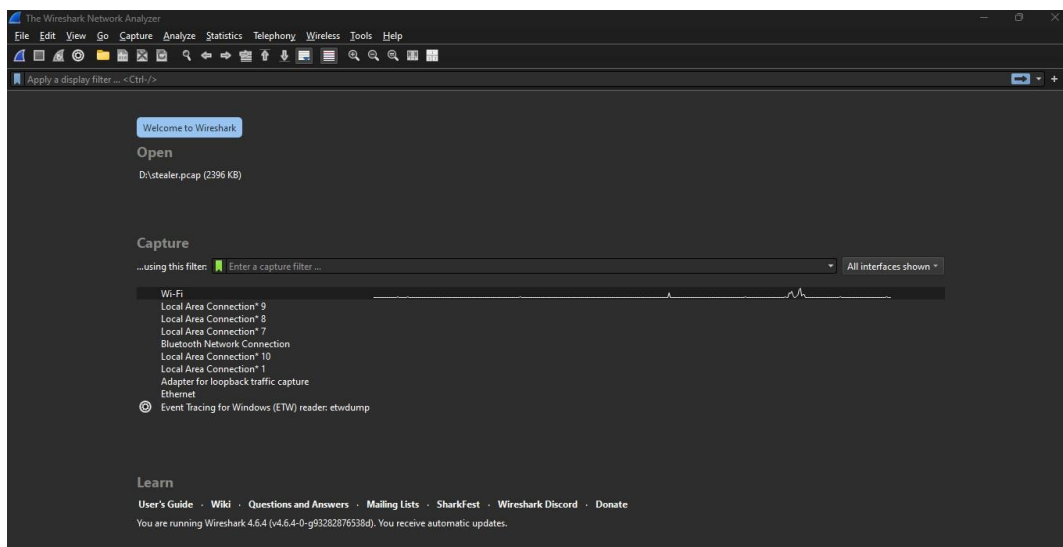


Figure 4.3.

4.4. Select .pcap File

Opened Wireshark and chose the downloaded .pcap file.

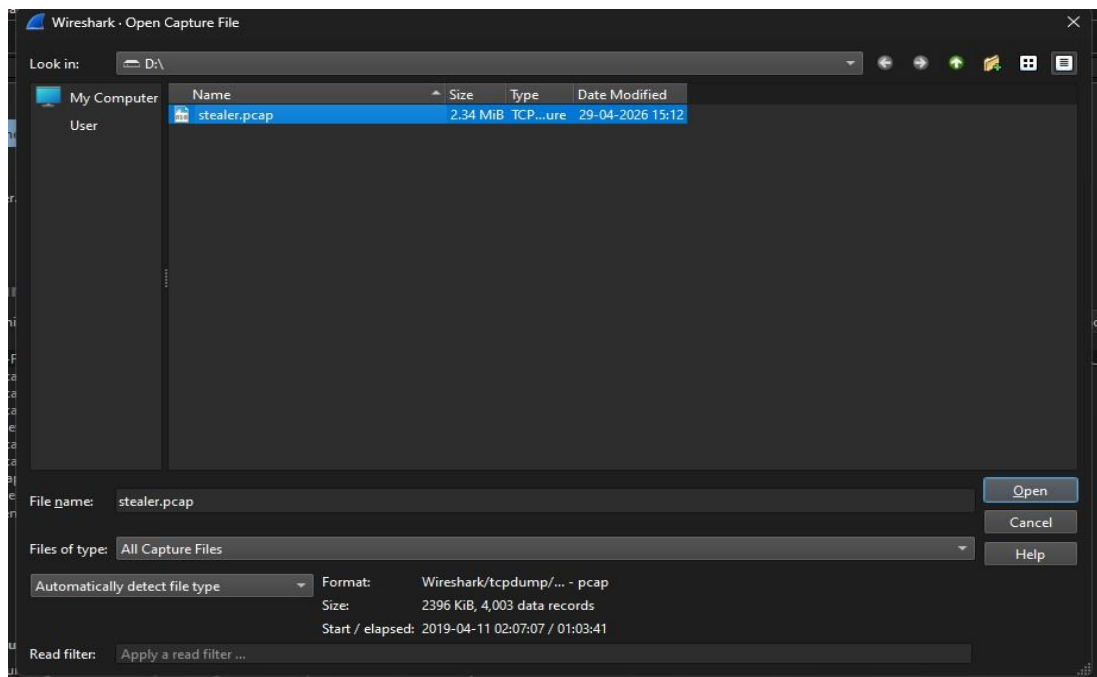


Figure 4.4.

5. Evidence Provided

- pcap file containing captured network traffic
- simulated victim machine traffic
- protocols observed – HTTP, TCP, SNMP, DNS.

6. Detailed Analysis – HawkEye Lab Questions

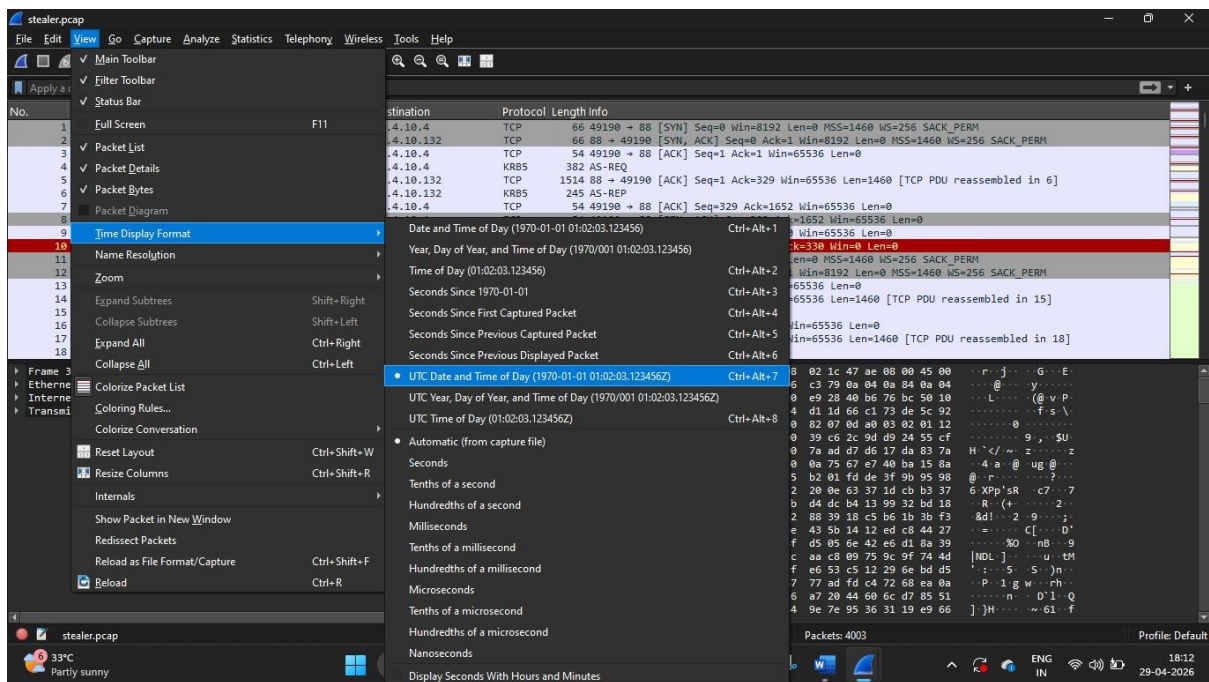
Q1. How many packets does the capture have?

4003 packets in total as seen in the snapshot.

4001	2019-04-10 21:40:48.6022832	10.4.10.4	SNMP	182	10c1	REQUEST PDU::UDP_GET_REFERRALS	
4002	2019-04-10 21:40:48.6023912	10.4.10.4	SNMP	278	10c1	Response PDU::UDP_GET_REFERRALS	
4003	2019-04-10 21:40:48.6096337	10.4.10.4	TCP	54	49228	→ 445 [ACK] Seq=3836 Ack=1073 Win=64512 Len=0	

Q2. At what time was the first packet captured (UTC)?

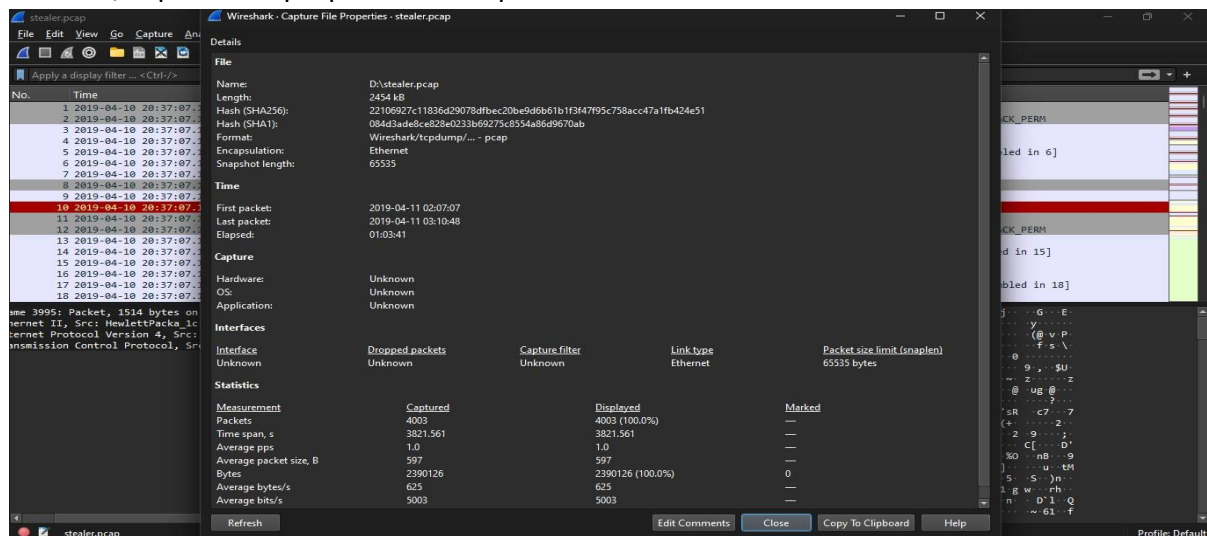
2019-04-10 20:37, sort time display format as UTC.



No.	Time	Source	Destination	Protocol
1	2019-04-10 20:37:07.129730Z	10.4.10.132	10.4.10.4	TCP
2	2019-04-10 20:37:07.129811Z	10.4.10.4	10.4.10.132	TCP

Q3. What is the duration of the capture?

01:03:41, capture file properties -> elapsed time.

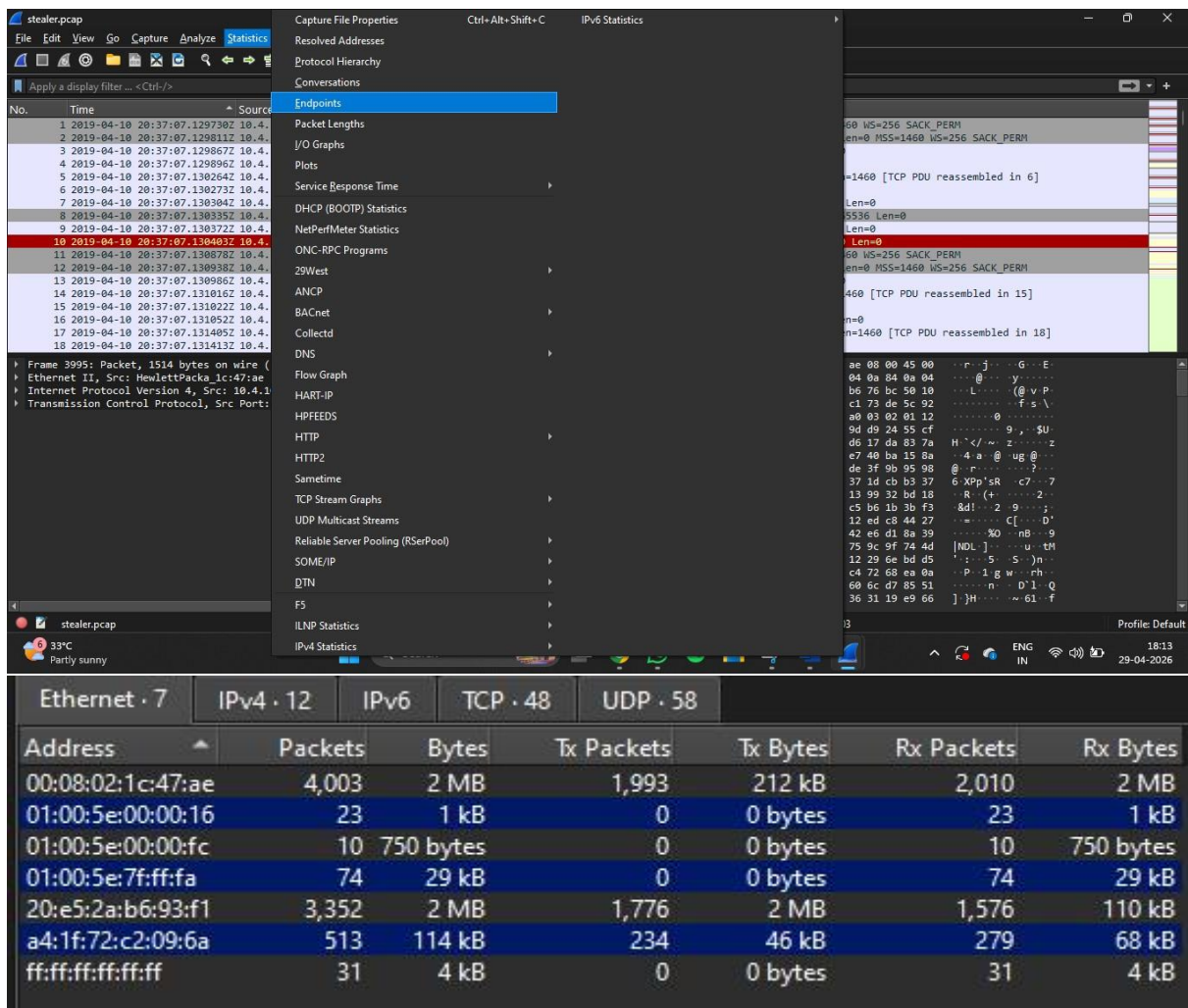


Elapsed: 01:03:41

Q4. What is the most active computer at the link level?

00:08:02:1c:47:ae, Statistics -> Endpoints -> Ethernet

MAC address of most active computer with 1993 transmitted packets and 2010 received packets in the section. The link layer has the data regarding MAC addresses.



Q5. Manufacturer of the NIC of the most active system at the link level?

Hewlett-Packard

Statistics -> Endpoints -> Ethernet

The manufacturer of the network interface card found on applying the filter of that particular MAC Address from the endpoints, which then shows the manufacturer under Ethernet.

The image shows a Wireshark network traffic analysis interface. The 'Endpoint Settings' dialog is open, displaying a table of endpoints with columns for Address, Packets, and Bytes. A context menu is open over the table, showing options like 'Apply as Filter', 'Prepare as Filter', 'Find', 'Colorize', 'Copy Endpoint table', and 'Resize all columns to content'. The main packet list shows a series of network packets, and the packet details pane shows the structure of a selected packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data of the packet.

Q6. Where is the headquarter of the company that manufactured the NIC of the most active computer at the link level?

Palo Alto from the Google search engine.

Google

hewlett packard headquarters

AI Mode

All

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★ AI Overview

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🔊

HP Inc.'s global headquarters is located at 1501 Page Mill Road, Palo Alto, California, 94304, USA. Following the 2015 split, HP Inc. (personal systems, printing) retained the Palo Alto location, while Hewlett Packard Enterprise (HPE) moved its headquarters to Spring, Texas.

HP +3

Q7. The organization works with private addressing and netmask /24. How many computers in the organization are involved in the capture?

3, Statistics -> Endpoints -> IPV4

IP Addresses with 10.4.10.2, 10.4.10.4, 10.4.10.132 are the computers used, 10.4.10.255 is a broadcast address so it is not considered.

Ethernet · 7	IPv4 · 12	IPv6	TCP · 48	UDP · 58			
Address ▲	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
10.4.10.2	42	5 kB	0	0 bytes	42	5 kB	
10.4.10.4	513	114 kB	234	46 kB	279	68 kB	
10.4.10.132	4,003	2 MB	1,993	212 kB	2,010	2 MB	
10.4.10.255	30	3 kB	0	0 bytes	30	3 kB	
23.229.162.69	280	39 kB	161	13 kB	119	26 kB	
66.171.248.178	63	5 kB	28	3 kB	35	2 kB	
216.58.193.131	20	8 kB	11	6 kB	9	3 kB	
217.182.138.150	2,947	2 MB	1,576	2 MB	1,371	74 kB	
224.0.0.22	23	1 kB	0	0 bytes	23	1 kB	
224.0.0.252	10	750 bytes	0	0 bytes	10	750 bytes	
239.255.255.250	74	29 kB	0	0 bytes	74	29 kB	
255.255.255.255	1	342 bytes	0	0 bytes	1	342 bytes	

Q8. What is the name of the most active computer at the network level?

Beijing-5cd1-PC

1. Statistics -> Endpoints -> IPV4
2. DHCP filter

At the network level, we look for IP address with the highest amount of data transfer, that is 10.4.10.132. On applying the dhcp filter, we get the host name of the respective IP Address.

Ethernet · 7	IPv4 · 12	IPv6	TCP · 48	UDP · 58		
Address 	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
10.4.10.2	42	5 kB	0	0 bytes	42	5 kB
10.4.10.4	513	114 kB	234	46 kB	279	68 kB
10.4.10.132	4,003	2 MB	1,993	212 kB	2,010	2 MB
10.4.10.255	30	3 kB	0	0 bytes	30	3 kB

stealer.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcpc

No.	Time	Source	Destination	Protocol	Length	Info
3263	2019-04-10 20:47:56.3246012	10.4.10.132	255.255.255.255	DHCP	342	DHCP Inform - Transaction ID 0xc0361803
3264	2019-04-10 20:47:56.3250652	10.4.10.4	10.4.10.132	DHCP	342	DHCP ACK - Transaction ID 0xc0361803

Frame 3264: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on Ethernet II, Src: Dell_c2:09:6a (a4:1f:72:c2:09:6a), Dst: HewlettPacka_1c:47:ae (08:00:02:1c:47:ae)

Destination: HewlettPacka_1c:47:ae (08:00:02:1c:47:ae)

Source: Dell_c2:09:6a (a4:1f:72:c2:09:6a)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 10.4.10.4, Dst: 10.4.10.132

User Datagram Protocol, Src Port: 67, Dst Port: 68

Dynamic Host Configuration Protocol (ACK)

Message type: Boot Reply (2)

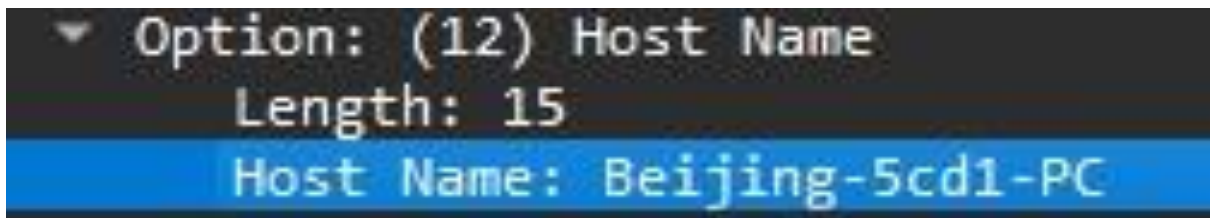
Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0xc0361803

Seconds elapsed: 0



Q9. What is the IP of the organization's DNS server?

10.4.10.4

On applying DNS filter, we can see that the source IP Address is 10.4.10.4 which is the DNS server answering a query from the client.

stealer.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
119	2019-04-10 20:37:33.3783907	10.4.10.4	10.4.10.132	DNS	102	Standard query response 0x3ee5 No such name SRV ldap, tcp.pizzajukebox-DC.pizzajukebox.com
174	2019-04-10 20:37:33.9116512	10.4.10.132	10.4.10.4	DNS	76	Standard query 0x8701 A dns.msftncsi.com

Q10. What domain is the victim asking about in packet 204?

proforma-invoices.com

This is suspicious because the domain is used to host a malicious executable file, which is a common technique used by attackers to distribute malware.

204	2019-04-10	20:37:53.791017Z	10.4.10.132	10.4.10.4	DNS	81	Standard query 0xa002 A proforma-invoices.com
205	2019-04-10	20:37:54.521557Z	10.4.10.132	10.4.10.255	NBNS	92	Name query NB WPA<00>
206	2019-04-10	20:37:54.577019Z	10.4.10.4	10.4.10.132	DNS	97	Standard query response 0xa002 A proforma-invoices.com A 217.182.138.150
207	2019-04-10	20:37:54.588941Z	10.4.10.132	217.182.138.150	TCP	66	49204 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
208	2019-04-10	20:37:54.726874Z	217.182.138.150	10.4.10.132	TCP	58	80 → 49204 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
209	2019-04-10	20:37:54.727037Z	10.4.10.132	217.182.138.150	TCP	54	49204 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
210	2019-04-10	20:37:54.727262Z	10.4.10.132	217.182.138.150	HTTP	392	GET /proforma/tkraw_Protected99.exe HTTP/1.1
211	2019-04-10	20:37:54.727339Z	217.182.138.150	10.4.10.132	TCP	54	80 → 49204 [ACK] Seq=1 Ack=339 Win=64240 Len=0
212	2019-04-10	20:37:54.864236Z	217.182.138.150	10.4.10.132	TCP	286	80 → 49204 [PSH, ACK] Seq=1 Ack=339 Win=64240 Len=232 [TCP PDU reassembled in 3155]

.....0.....	= IG bit: Individual address (unicast)	0000	a4 1f 72 c2 09 6a 00 08 02 1c 47 ae 00 00 45 00	G.....E.....
Source: HewlettPacka_1c:47:ae (00:08:02:1c:47:ae)		0010	00 43 01 9f 00 00 00 11 10 7c 0a 0a 0a 04 0a 04	S.....
.....0.....	= IG bit: Globally unique address (factory default)	0020	0a 04 d5 86 00 35 00 2f 7e d2 a0 02 01 00 00 01	5 / w.....
.....0.....	= IG bit: Individual address (unicast)	0030	00 00 00 00 00 00 11 70 72 6f 66 6f 72 6d 61 2d	p roforma-
Type: IPv4 (0x0800)		0040	69 6e 76 6f 69 63 65 73 03 63 6f 6d 00 00 01 00	invoices .com....
[Stream index: 0]		0050	01	

Internet Protocol Version 4, Src: 10.4.10.132, Dst: 10.4.10.4

User Datagram Protocol, Src Port: 54662, Dst Port: 53

Domain Name System (query)

Transaction ID: 0xa002

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

proforma-invoices.com: type A, class IN

[Response In: 206]

Queries

proforma-invoices.com: type A, class IN

Q11. What is the IP of the domain in the previous question?

217.182.138.150

The IP Address of the domain name query is given a query response in the answers section for port 206.

206	2019-04-10	20:37:54.577019Z	10.4.10.4	10.4.10.132	DNS	97	Standard query response 0xa002 A proforma-invoices.com A 217.182.138.150
3159	2019-04-10	20:38:15.672284Z	10.4.10.132	10.4.10.4	DNS	85	Standard query 0x3f59 A bot.whatismyipaddress.com
3160	2019-04-10	20:38:15.706148Z	10.4.10.4	10.4.10.132	DNS	101	Standard query response 0x3f59 A bot.whatismyipaddress.com A 66.171.248.178
3170	2019-04-10	20:38:15.832004Z	10.4.10.132	10.4.10.4	DNS	78	Standard query 0x3daa A macwinlogistics.in
3171	2019-04-10	20:38:15.912953Z	10.4.10.4	10.4.10.132	DNS	94	Standard query response 0x3daa A macwinlogistics.in A 23.229.162.69
3268	2019-04-10	20:47:58.641867Z	10.4.10.132	10.4.10.4	DNS	81	Standard query 0x65d6 A update.googleapis.com
3269	2019-04-10	20:47:58.675070Z	10.4.10.4	10.4.10.132	DNS	97	Standard query response 0x65d6 A update.googleapis.com A 216.58.193.131
3290	2019-04-10	20:48:20.054661Z	10.4.10.132	10.4.10.4	DNS	85	Standard query 0x3119 A bot.whatismyipaddress.com
3291	2019-04-10	20:48:20.077594Z	10.4.10.4	10.4.10.132	DNS	101	Standard query response 0x3119 A bot.whatismyipaddress.com A 66.171.248.178
3301	2019-04-10	20:48:20.203215Z	10.4.10.132	10.4.10.4	DNS	78	Standard query 0x50d8 A macwinlogistics.in
3302	2019-04-10	20:48:20.334494Z	10.4.10.4	10.4.10.132	DNS	94	Standard query response 0x50d8 A macwinlogistics.in A 23.229.162.69

.....0.....	= Answer authenticated: Answer/authority portion was not authentic	0000	00 00 02 1c 47 ae a4 1f 72 c2 09 6a 00 00 45 00	G.....E.....
.....0.....	= Non-authenticated data: Unacceptable	0010	00 53 09 5f 40 00 80 11 c8 ab 8a 04 0a 04 0a 04	S.....
.....0.....	= Reply code: No error (0)	0020	0a 84 00 35 d5 86 00 3f 5f d4 a0 02 01 80 00 01	5 / 2.....
Questions: 1		0030	00 01 00 00 00 00 11 70 72 6f 66 6f 72 6d 61 2d	p roforma-
Answer RRs: 1		0040	69 6e 76 6f 69 63 65 73 03 63 6f 6d 00 00 01 00	invoices .com....
Authority RRs: 0		0050	01 c0 0c 00 01 00 01 00 00 3e 00 04 d9 b6 8a	>.....
Additional RRs: 0		0060	96	

proforma-invoices.com: type A, class IN

Name: proforma-invoices.com

[Name Length: 21]

[Label Count: 2]

Type: A (1) (Host Address)

Class: IN (0x0001)

Answers

proforma-invoices.com: type A, class IN, addr 217.182.138.150

[Request In: 204]

[Time: 786.002000 milliseconds]

Answers

proforma-invoices.com: type A, class IN, addr 217.182.138.150

Q12. Indicate the country to which the IP in the previous section belongs.

France, using the iplocation finder.

IP Location Finder

IPv4, IPv6 or Domain Name

IP Lookup

Enter a IPv4, IPv6 or Domain name into the input box above, and we'll locate its IP location.

217.182.138.150



Hide this IP Address

Here are the results from a few Geolocation providers. Is the data shown below not accurate enough? Please read [geolocation accuracy info](#) to learn why.

Do you have a problem with IP location lookup? [Report a problem.](#)

Geolocation data from		IP2Location	Product: DB6, 2026-4-15
IP ADDRESS:	217.182.138.150	ISP:	OVH SAS
COUNTRY:	France	ORGANIZATION:	Not available
REGION:	Hauts-de-France	LATITUDE:	50.6937
CITY:	Roubaix	LONGITUDE:	3.1744

Q13. What operating system does the victim's computer run? Windows

NT 6.1

http can be used to isolate web traffic and give details regarding OS, web browsers, thus showing it in the User-Agent section.

Wireshark packet capture showing an HTTP GET request from 217.182.138.150 to 10.4.10.132. The packet details show the User-Agent as 'Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30619; .NET CLR 3.0.45062.5; .NET CLR 3.5.50724; .NET CLR 4.0.30319.1)'.

```
Hypertext Transfer Protocol
  GET /proforma/tkraw_Protected99.exe HTTP/1.1\r\n
  Accept: */*\r\n
  Accept-Encoding: gzip, deflate\r\n
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30619; .NET CLR 3.0.45062.5; .NET CLR 3.5.50724; .NET CLR 4.0.30319.1)\r\n
  Host: proforma-invoices.com\r\n
  Connection: Keep-Alive\r\n
```

Q14. What is the name of the malicious file downloaded by the accountant?

http://proforma-invoices.com/proforma/tkraw_Protected99.exe

The .exe file indicates presence of malware as the attackers often disguise malicious files as legitimate applications.

```

Hypertext Transfer Protocol
  GET /proforma/tkraw_Protected99.exe HTTP/1.1\r\n
  Accept: */*\r\n
  Accept-Encoding: gzip, deflate\r\n
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0;
  Host: proforma-invoices.com\r\n
  Connection: Keep-Alive\r\n
  \r\n
  [Response in frame: 3155]
  [Full request URI: http://proforma-invoices.com/proforma/tkraw_Protected99.exe]

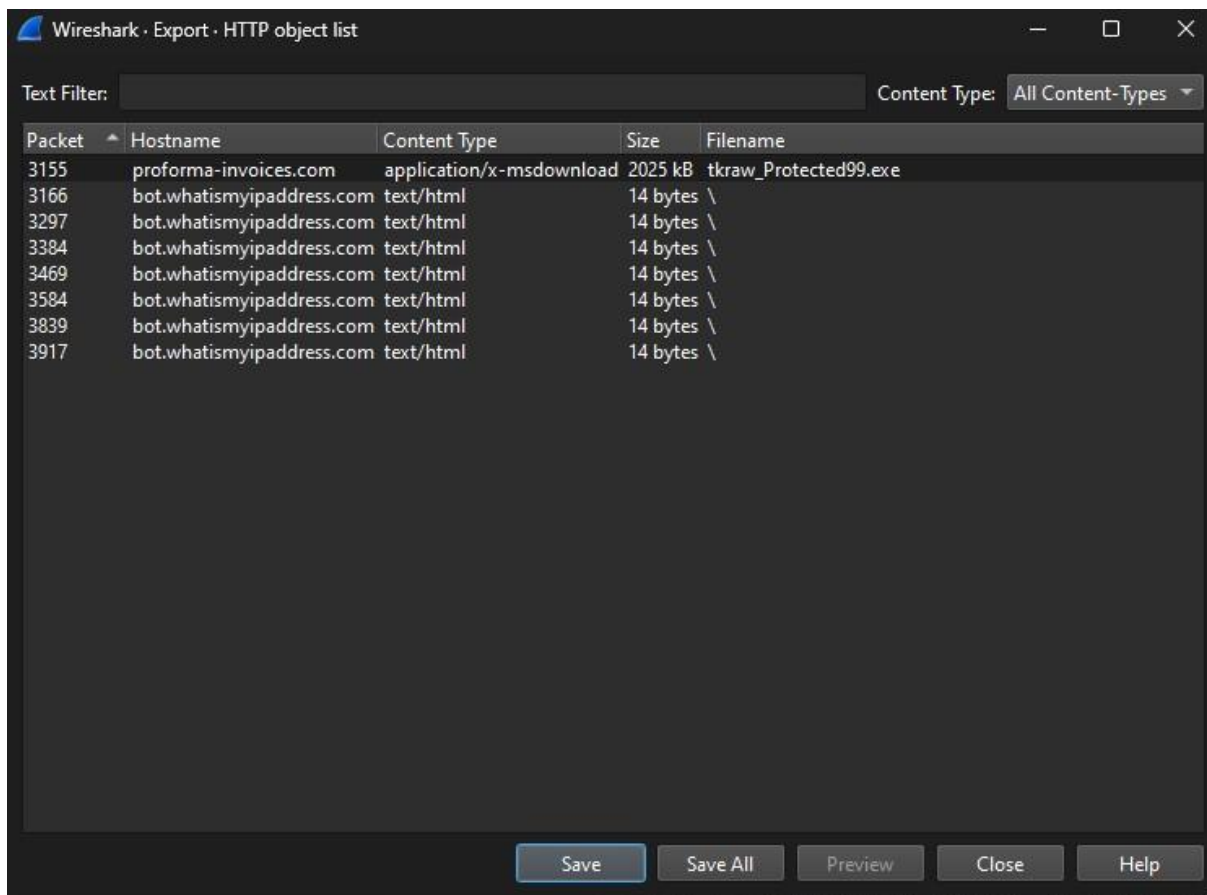
```

Q15. What is the md5 hash of the downloaded file?

MD5 - 71826BA081E303866CE2A2534491A2F7

1. Download file.
2. Save in Users>User> tkraw_Protected99.exe
3. Run command in Windows Powershell for md5 hash of the file:

PS C:\Users\User> get-filehash -Algorithm md5 .\tkraw_Protected99.exe




```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\User> get-filehash -Algorithm md5 .\tkraw_Protected99.exe

Algorithm      Hash
-----
MD5             71826BA081E303866CE2A2534491A2F7

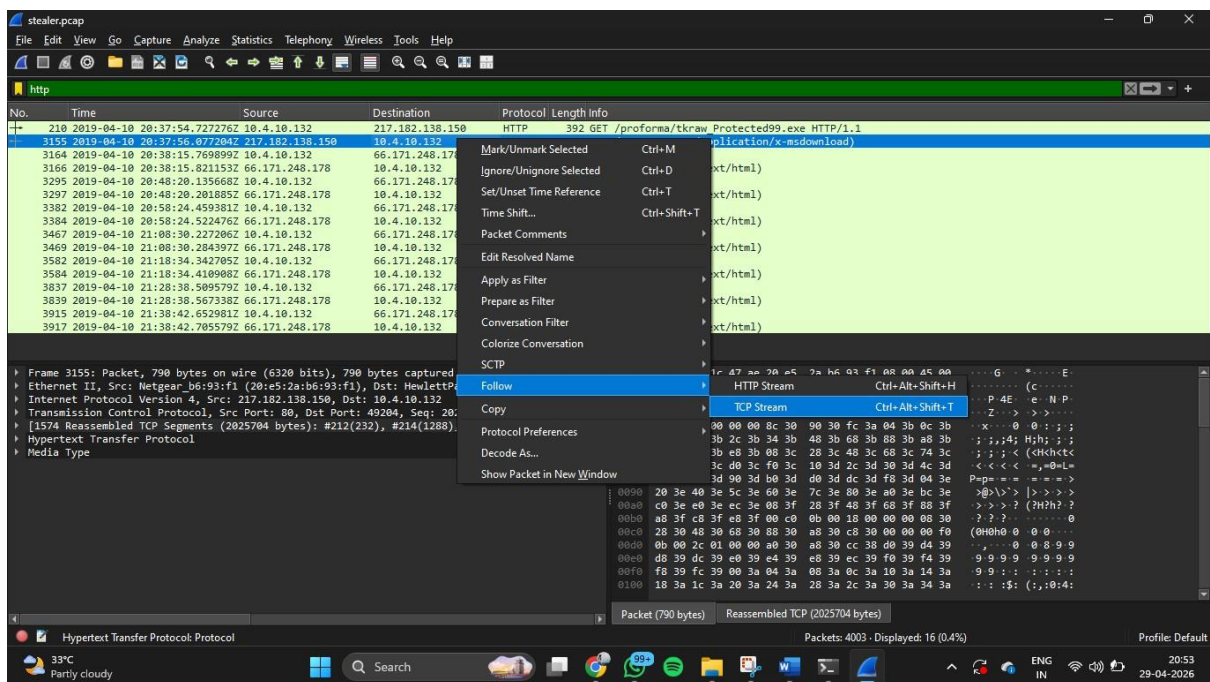
Path
----
C:\Users\User\tkraw_Protected...

PS C:\Users\User>

```

Q16. What software runs the webserver that hosts the malware?

LiteSpeed is the software used. We can track it by following the TCP stream of the malware file used.



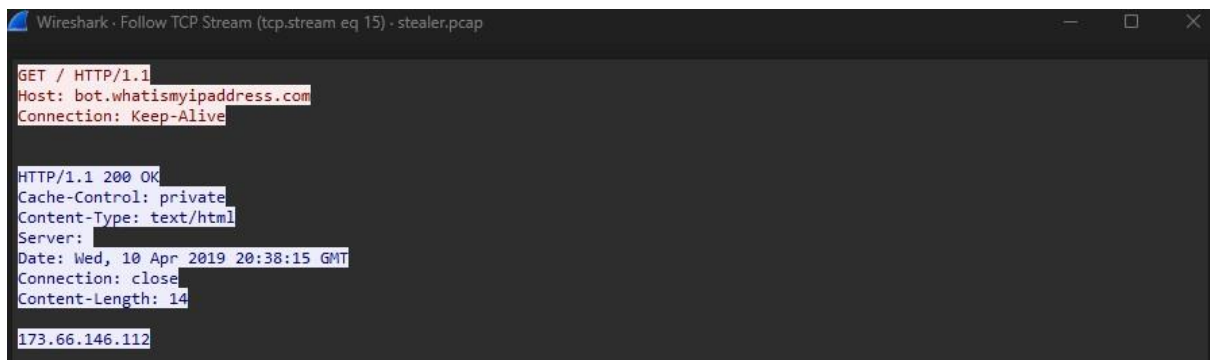
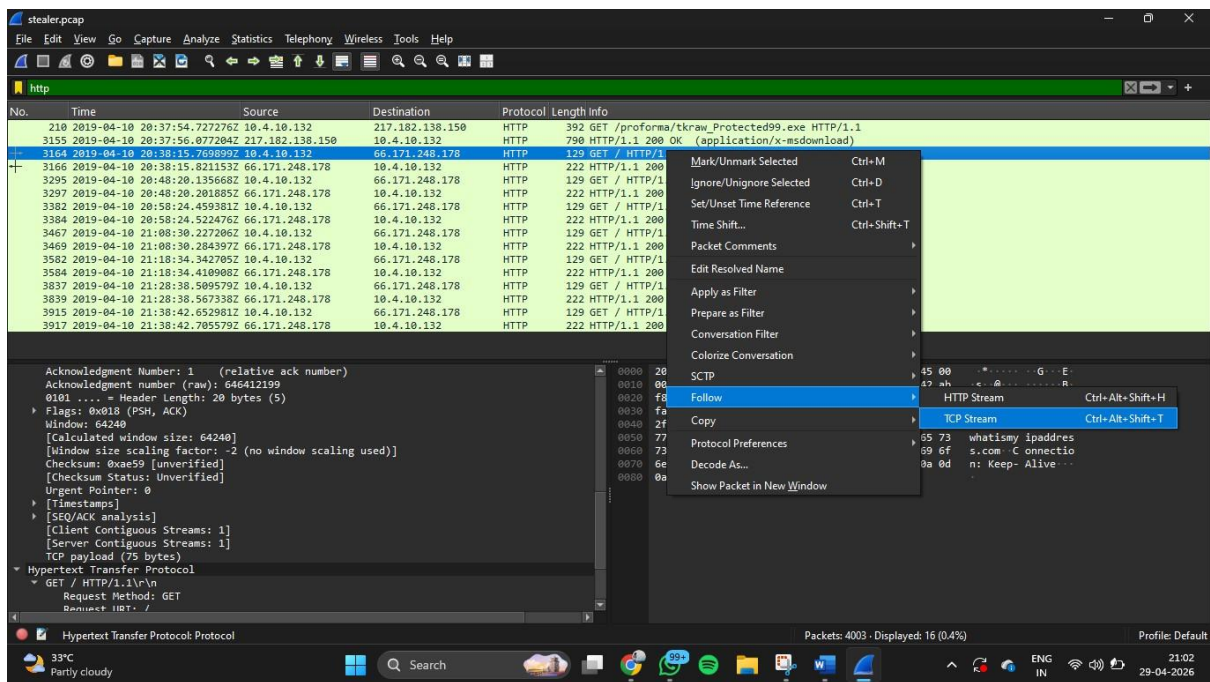
```

HTTP/1.1 200 OK
Last-Modified: Wed, 10 Apr 2019 04:44:31 GMT
Content-Type: application/x-msdownload
Content-Length: 2025472
Accept-Ranges: bytes
Date: Wed, 10 Apr 2019 20:37:54 GMT
Server: LiteSpeed
Connection: Keep-Alive

```

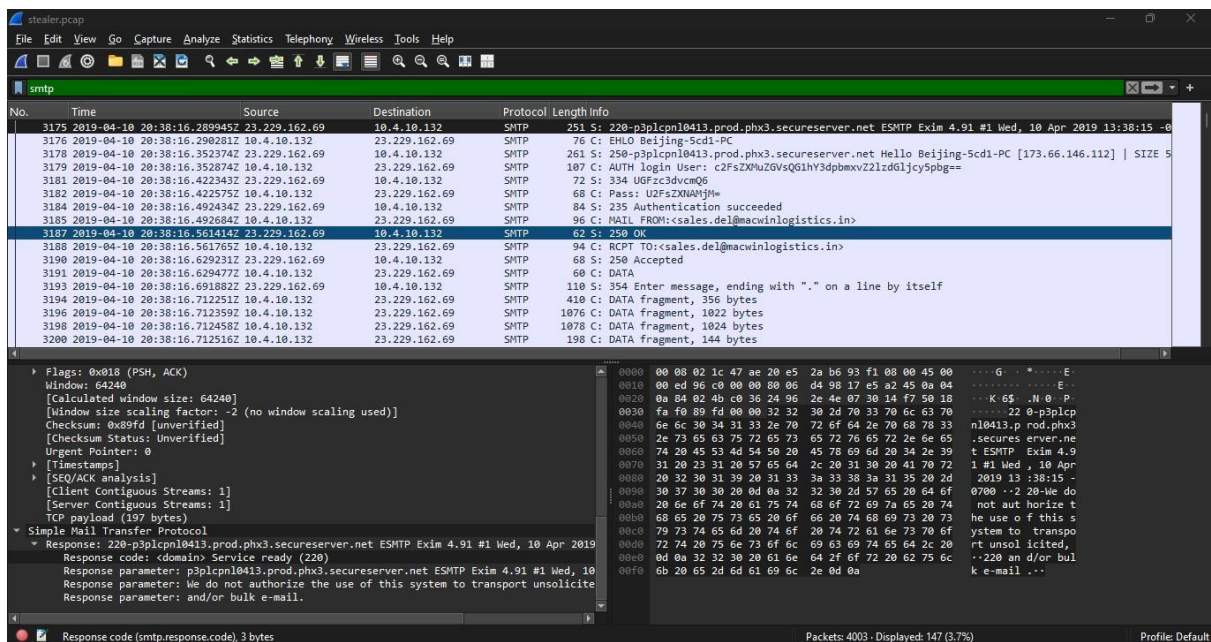
Q17. What is the public IP of the victim's computer?

173.66.146.112. On following the TCP stream of the GET request asking for the its IP address we find the victim's public IP address which helps in tracing external communication.



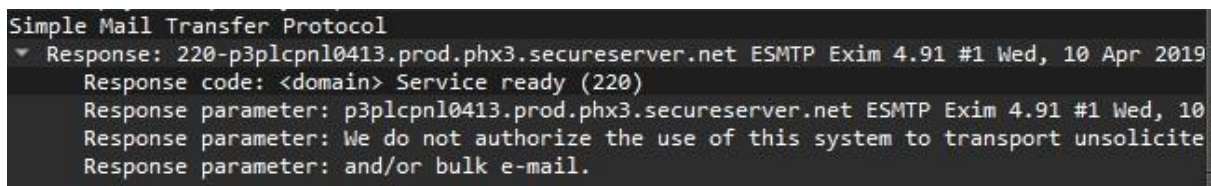
Q18. In which country is the email server to which the stolen information is sent?

United States, on applying the SMTP filter, we get the source IP as 23.229.162.69. On using the iplocation finder we get the geolocation.



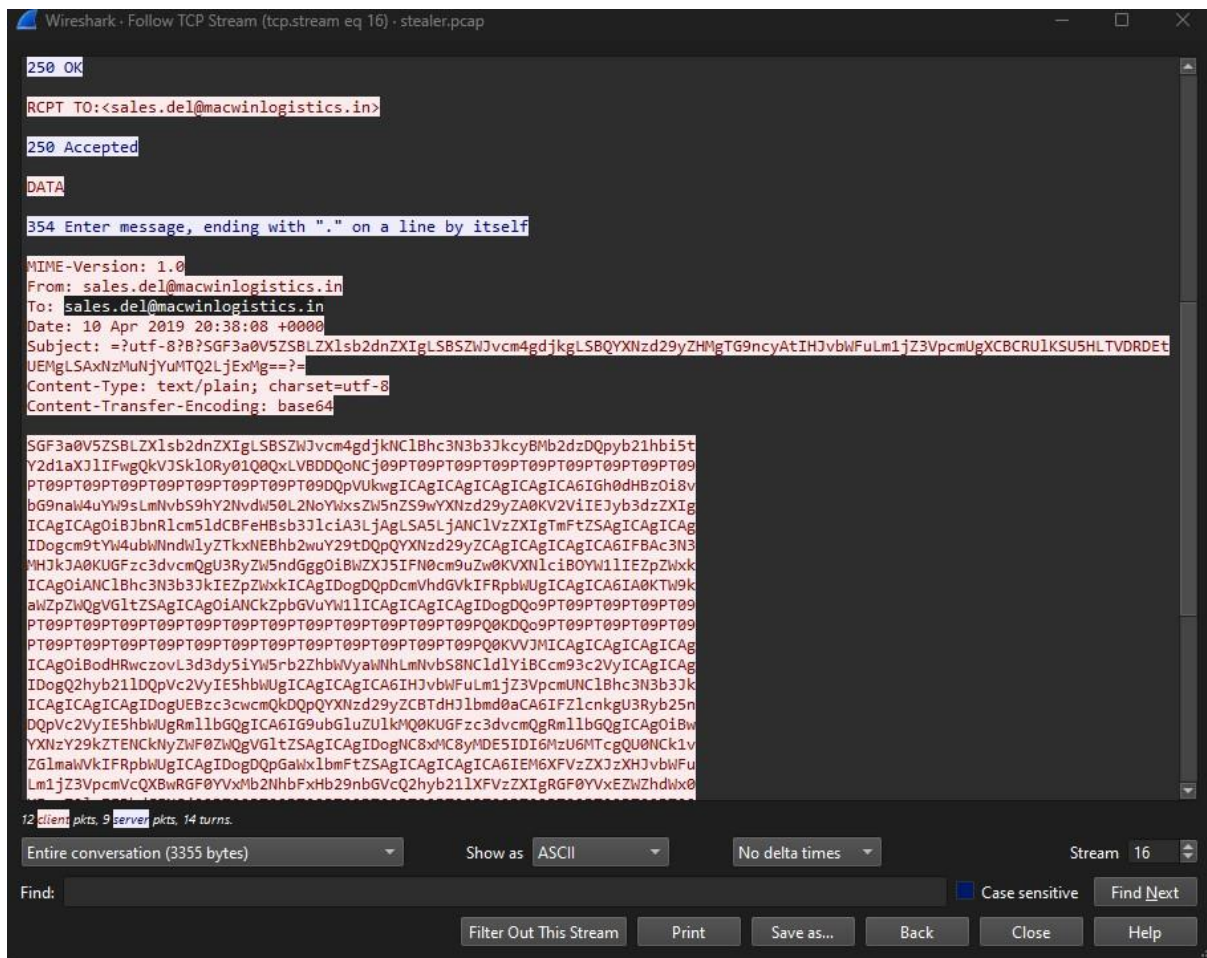
Q19. Analyzing the first extraction of information. What software runs the email server to which the stolen data is sent?

Exim 4.91, on applying the SMTP filter, we get the source IP as 23.229.162.69. And in the SMTP section it shows the server used.



Q20. To which email account is the stolen information sent?

sales.del@macwinlogistics.in, on following the TCP stream, it gives us information about data exfiltration which is a known attacker technique to steal data and share it with them.



Q21. What is the password used by the malware to send the email?

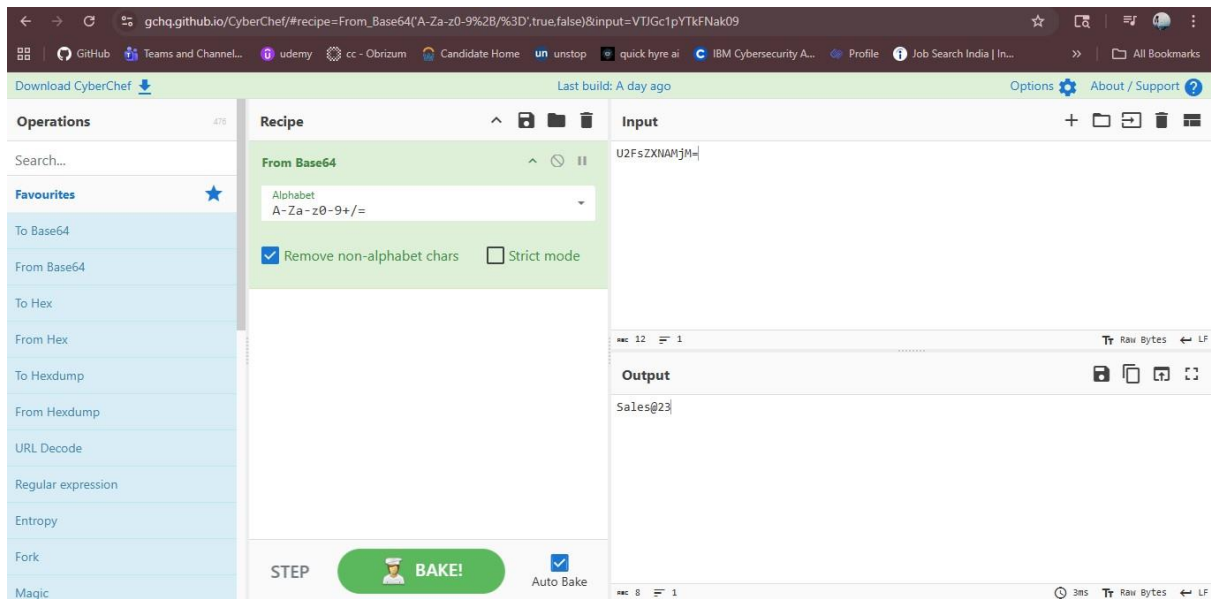
Sales@123.

We get the Base64 encoded string from the SMTP authentication packet, on using CyberChef to decode it, we get the plaintext that was used as the password.

No.	Time	Source	Destination	Protocol	Length	Info
3175	2019-04-10 20:38:16.2899457	23.229.162.69	10.4.10.132	SMTP	251	S: 220-p3lpcnl0413.prod.phx3.secureserver.net ESMTX Exim 4.91 #1 Wed, 10 Apr 2019 13:38:15 -0
3176	2019-04-10 20:38:16.2902812	10.4.10.132	23.229.162.69	SMTP	76	C: EHLO Beijing-5cd1-PC
3178	2019-04-10 20:38:16.3523742	23.229.162.69	10.4.10.132	SMTP	261	S: 250-p3lpcnl0413.prod.phx3.secureserver.net Hello Beijing-5cd1-PC [173.66.146.112] SIZE 5
3179	2019-04-10 20:38:16.3528742	10.4.10.132	23.229.162.69	SMTP	107	C: AUTH login User: c2FsZXNzZGVsQGlhY3dpbmVZl2d6ljcy5pbG==
3181	2019-04-10 20:38:16.4223432	23.229.162.69	10.4.10.132	SMTP	72	S: 334 UGFzc3dvcmQ=
3182	2019-04-10 20:38:16.4225762	10.4.10.132	23.229.162.69	SMTP	68	C: PASS: U2FsZXNzZGVsQGlhY3dpbmVZl2d6ljcy5pbG==
3184	2019-04-10 20:38:16.4924342	23.229.162.69	10.4.10.132	SMTP	84	S: 235 Authentication succeeded
3185	2019-04-10 20:38:16.4926842	10.4.10.132	23.229.162.69	SMTP	96	C: MAIL FROM:<sales.del@macwinlogistics.in>
3187	2019-04-10 20:38:16.5614142	23.229.162.69	10.4.10.132	SMTP	62	S: 250 OK
3188	2019-04-10 20:38:16.5617652	10.4.10.132	23.229.162.69	SMTP	94	C: RCPT TO:<sales.del@macwinlogistics.in>
3190	2019-04-10 20:38:16.6292312	23.229.162.69	10.4.10.132	SMTP	68	S: 250 Accepted
3191	2019-04-10 20:38:16.6294772	10.4.10.132	23.229.162.69	SMTP	60	C: DATA
3193	2019-04-10 20:38:16.6918822	23.229.162.69	10.4.10.132	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
3194	2019-04-10 20:38:16.7125212	10.4.10.132	23.229.162.69	SMTP	410	C: DATA fragment, 356 bytes
3196	2019-04-10 20:38:16.7123592	10.4.10.132	23.229.162.69	SMTP	1076	C: DATA fragment, 1022 bytes
3198	2019-04-10 20:38:16.7124582	10.4.10.132	23.229.162.69	SMTP	1078	C: DATA fragment, 1024 bytes
3200	2019-04-10 20:38:16.7125162	10.4.10.132	23.229.162.69	SMTP	198	C: DATA fragment, 144 bytes

Simple Mail Transfer Protocol

Password: U2FsZXNzZGVsQGlhY3dpbmVZl2d6ljcy5pbG==



Q22. Which malware variant exfiltrated the data?

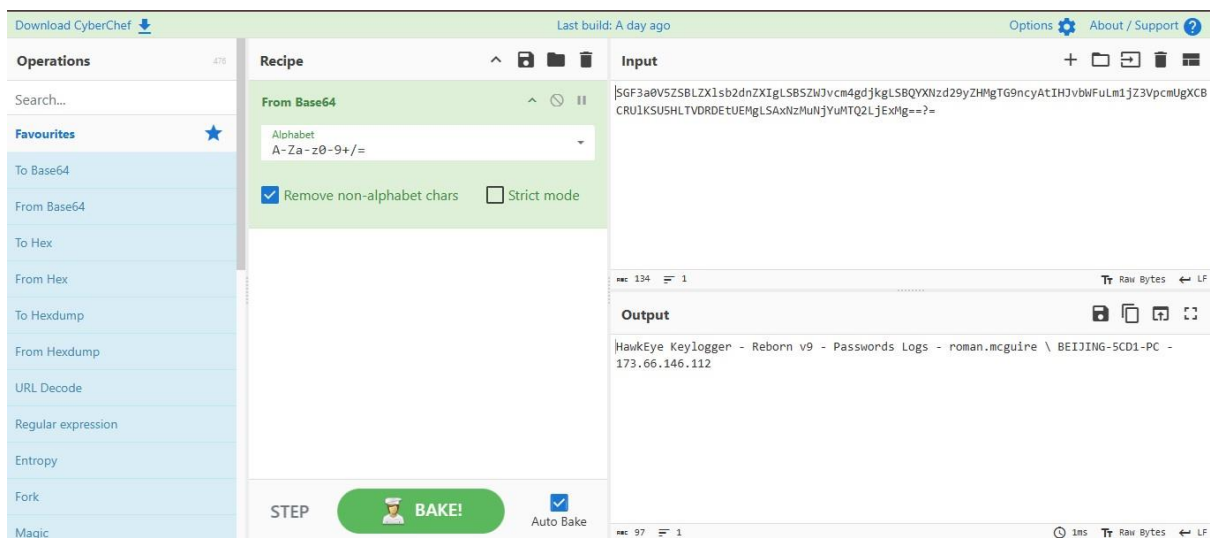
Reborn v9

Decoding the subject using CyberChef gives us the malware variant used.

```

MIME-Version: 1.0
From: sales.del@macwinlogistics.in
To: sales.del@macwinlogistics.in
Date: 10 Apr 2019 20:38:08 +0000
Subject: =?utf-8?B?SGF3a0V5ZSBLZX1sb2dnZXIgL SB5ZWJvc m4gdjkgLSBQYXNzd29yZHMgTG9ncyAtIHJvbWVudm1jZ3Vpc mUgXCB CRU1KSUSHLTVDRDEtUEMgLSAxNzMuNjYuMTQ2LjExMg==?=
Content-Type: text/plain; charset=utf-8
Content-Transfer-Encoding: base64

```



Q23. What are the bankofamerica access credentials? (username:password)

roman.mcguire:P@ssw0rd\$, decoded using CyberChef.

No.	Time	Source	Destination	Protocol	Length	Info
3321	2019-04-10 20:48:20.9828942	23.229.162.69	10.4.10.132	SMTP	68 S	250 Accepted
3322	2019-04-10 20:48:20.9836672	10.4.10.132	23.229.162.69	SMTP	60 C	DATA
3324	2019-04-10 20:48:21.0473292	23.229.162.69	10.4.10.132	SMTP	110 S	354 Enter message, ending with "." on a line by itself
3325	2019-04-10 20:48:21.0476692	10.4.10.132	23.229.162.69	SMTP	410 C	DATA fragment, 356 bytes
3326	2019-04-10 20:48:21.0476962	10.4.10.132	23.229.162.69	SMTP	1076 C	DATA fragment, 182 bytes
3329	2019-04-10 20:48:21.0478952	10.4.10.132	23.229.162.69	SMTP	1078 C	DATA fragment, 1024 bytes
3330	2019-04-10 20:48:21.0478632	10.4.10.132	23.229.162.69	SMTP	198 C	DATA fragment, 144 bytes
3333	2019-04-10 20:48:21.0481222	10.4.10.132	23.229.162.69	SMTP	56 C	DATA fragment, 2 bytes
3335	2019-04-10 20:48:21.0481872	10.4.10.132	23.229.162.69	SMTP/L	59	from: sales.del@macwinlogistics.in, subject: =?utf-8?SGF3a0V5ZSBLZXlsbn2XIGlS8S2WJvcmgdkgI=
3337	2019-04-10 20:48:21.1946172	23.229.162.69	10.4.10.132	SMTP	82	250 OK 1E=IHEK9?00CVA=N8
3343	2019-04-10 20:49:57.5547673	23.229.162.69	10.4.10.132	SMTP	121	421 p3plcn10413.prod.phx3.secureserver.net lost input connection
3393	2019-04-10 20:58:24.7553572	23.229.162.69	10.4.10.132	SMTP	251 S	250-3p3plcn10413.prod.phx3.secureserver.net ESMTP Exim 4.91 #1 Wed, 10 Apr 2019 13:58:24 -0
3394	2019-04-10 20:58:24.7596862	10.4.10.132	23.229.162.69	SMTP	76 C	EHLO Beijing-5cd1-PC
3396	2019-04-10 20:58:24.8237102	23.229.162.69	10.4.10.132	SMTP	261 S	250-3p3plcn10413.prod.phx3.secureserver.net Hello Beijing-5cd1-PC [173.66.146.112] SIZE 5
3397	2019-04-10 20:58:24.8239302	10.4.10.132	23.229.162.69	SMTP	107 C	AUTH login User: c2FSZuXwZGVsQ5lH3dpbmvcZ2l2dGJcySpbg==
3399	2019-04-10 20:58:24.8938812	23.229.162.69	10.4.10.132	SMTP	72 S	334 UFGzC3dvcmQ6
3400	2019-04-10 20:58:24.8941162	10.4.10.132	23.229.162.69	SMTP	68 C	Pass: U2fS3ZXNjAjlH=

7. Indicators of Compromise (IoCs)

1. **Malicious Domain**
proforma-invoices.com was the domain used by the victim system to download a suspicious file.
2. **Malicious IP address**
217.182.138.150, extracted by investigating the malicious domain, used for communication.
3. **Victim Machine IP**
10.4.10.132, internal system initiating suspicious traffic with large amount of data transfer and malware download.
4. **Malicious File Name**
tkraw_Protected99.exe, .exe indicates executable file indicating malware. These IoCs can be used to create detection rules in SIEM systems to prevent similar attacks.

8. Attack Timeline

1. Victim IP – 10.4.10.132 attempts to access a suspicious domain, **proformainvoices.com**.
2. DNS query sent to DNS server- 10.4.10.4, receives **IP address 217.182.138.150**.
3. Victim connects to 217.182.138.150 over unencrypted **HTTP** rather than HTTPS.
4. Suspicious .exe file, **tkraw_Protected99.exe** downloaded by the victim.
5. Victim OS – **Windows NT 6.1**, more vulnerable since old version, indicates exploiting vulnerability.
6. Communication established with malicious IP.
7. Data exfiltration confirmed after observing SMTP traffic which indicates exfiltration through emails.
8. The malware establishes persistence and continues data exfiltration every 10 minutes indicating automated attacker control, thus confirming that the system is compromised.

9. Mitigation Strategies

1. **Block malicious content**
Domain: proforma-invoices.com
IP Address: 217.182.138.150
2. **Endpoint malware scanning and removal**
 - Perform infected host system scanning.
 - Remove detected malware tkraw_Protected99.exe.
 - Use updated Anti-malware/EDR tools.
3. **Enforce HTTPS**
 - Restrict downloading over HTTP.
 - Enforce HTTPS only downloading in web browsers.
 - Enforce blocking suspicious domains from web.

4. Continuous monitoring and alerts

- Automatically block suspicious domains.
- Detect repeated queries and investigate.
- Monitor Smtip traffic.

5. User awareness

Perform employee training on identifying suspicious links, phishing attacks.

6. Use updated software

Avoid using Windows 7 and switch to newer versions to ensure system is free of vulnerabilities.

10. Conclusion

The HawkEye Lab results in the need to constantly monitor the network, have secure communication channels, end user knowledge and awareness in order to prevent similar attacks. By identifying key Indicators of Compromise (IOCs) and reconstructing the attack timeline, it is evident that early detection and proper security controls could have mitigated the impact of the breach.